JRG WELDING CORP

Structural & Ornamental Welding · Aluminum · Steel · Stainless

[Company Address] · [Phone] · [Email] · Lic. [License #]

WORK PROPOSAL

Proposal No.: **JRG-2026- [###]**

Date: **July 1, 2026**

Valid for: **30 days**

PREPARED FOR

Client: **[Client Name]**
Address: **[Project Address]**
Phone / Email: **[Client Contact]**

PROJECT

Replacement of two (2) existing V-configuration wood roof rafter sections with structural aluminum members — like-for-like appearance, equal or greater strength.

1. EXISTING CONDITIONS

The residence is an A-frame type structure with exposed sloped roof rafters ("V" members) supporting the roof deck at the gable overhangs. The members scheduled for replacement are solid wood timbers with an actual cross-section of **3½" × 5½" (nominal 4×6)**. Two (2) V-sections (two pairs of sloped members meeting at the ridge) show deterioration and are to be removed and replaced.

2. SCOPE OF WORK

1. **Mobilization & protection:** Set up ladders/scaffolding as required; protect roofing, glazing, decking and landscaping in the work area.
2. **Temporary support:** Install temporary shoring to safely carry the roof load while each V-section is out.
3. **Demolition:** Carefully cut free and remove the two (2) existing wood V-rafter sections (four sloped members total, approx. length **[L ft]** each). Haul away and legally dispose of debris.
4. **Materials:** Furnish new **6061-T6 structural aluminum rectangular tube** supplied by **Eastern Metal Supply** (size per Section 3, final stock confirmation with EMS at time of order).
5. **Fabrication:** Cut members to exact field-verified lengths and angles; fabricate ridge and seat connections (welded aluminum brackets / plates); TIG/MIG welds per AWS D1.2 aluminum structural welding practice; drill for fasteners.
6. **Finish:** Finish aluminum in **[Dark Bronze powder coat / wood-grain finish / paint to match]** to visually match the existing dark-stained timbers.
7. **Installation:** Set new aluminum members in the exact position of the removed timbers using stainless-steel fasteners; isolate aluminum from dissimilar metals and treated wood with barrier tape/washers to prevent galvanic corrosion; re-attach existing roof connections; seal penetrations.
8. **Closeout:** Remove shoring, touch-up finish, final cleanup and walkthrough with owner.

3. MATERIAL SELECTION — STRENGTH MATCH

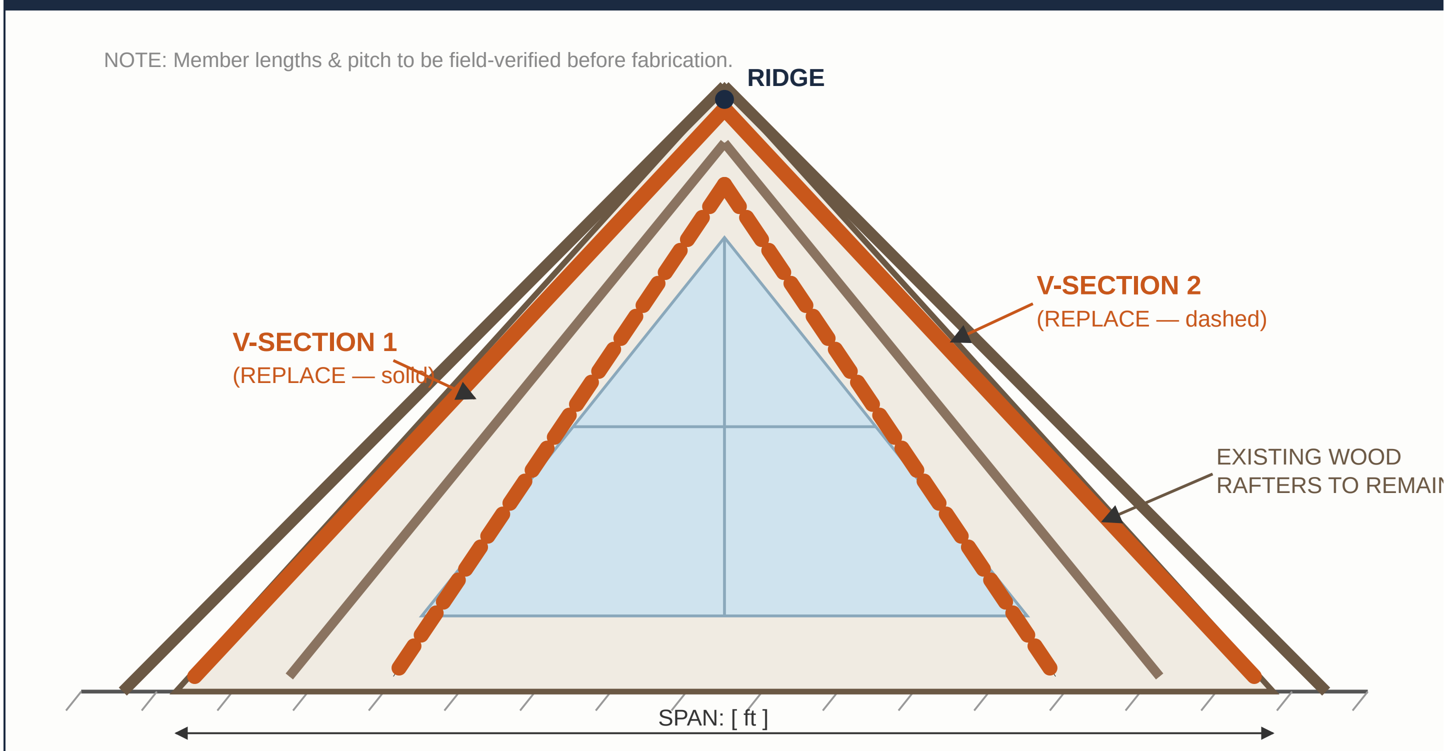
There is no aluminum tube produced in the exact 3½" × 5½" timber dimension, so the closest standard extrusions stocked by Eastern Metal Supply were evaluated. The table compares each candidate against the existing wood member (Southern Pine No. 2, F_b = 1,000 psi, E = 1.4×10^6 psi; allowable aluminum bending stress 19.5 ksi per Aluminum Design Manual, E = 10.1×10^6 psi):

Member	Weight (lb/ft)	Stiffness EI (lb-in²)	Stiffness vs. wood	Bending capacity vs. wood
Existing wood 4×6 (3.5"×5.5")	4.8	6.79×10^7	1.00×	1.00×
Aluminum 3"×5"×1/8" 6061-T6	2.3	6.76×10^7	0.99×	3.0×
Aluminum 3"×5"×3/16" 6061-T6	3.4	9.71×10^7	1.43×	4.2×
Aluminum 4"×6"×3/16" 6061-T6 RECOMMENDED	**4.2**	**1.84×10^8**	**2.7×**	**6.7×**
Aluminum 4"×6"×1/8" 6061-T6 (economy option)	2.9	1.27×10^8	1.9×	4.6×

Recommendation: 4" × 6" × 3/16" wall rectangular tube, alloy 6061-T6 (Eastern Metal Supply). It is the closest visual match to the existing timber (½" larger in each direction), provides **2.7× the stiffness and over 6× the bending strength** of the wood member it replaces, welds cleanly at 3/16" wall, and still weighs less than the timber. The 3/16" wall also gives comfortable margin even if the existing lumber were a stiffer species (e.g. Douglas Fir).

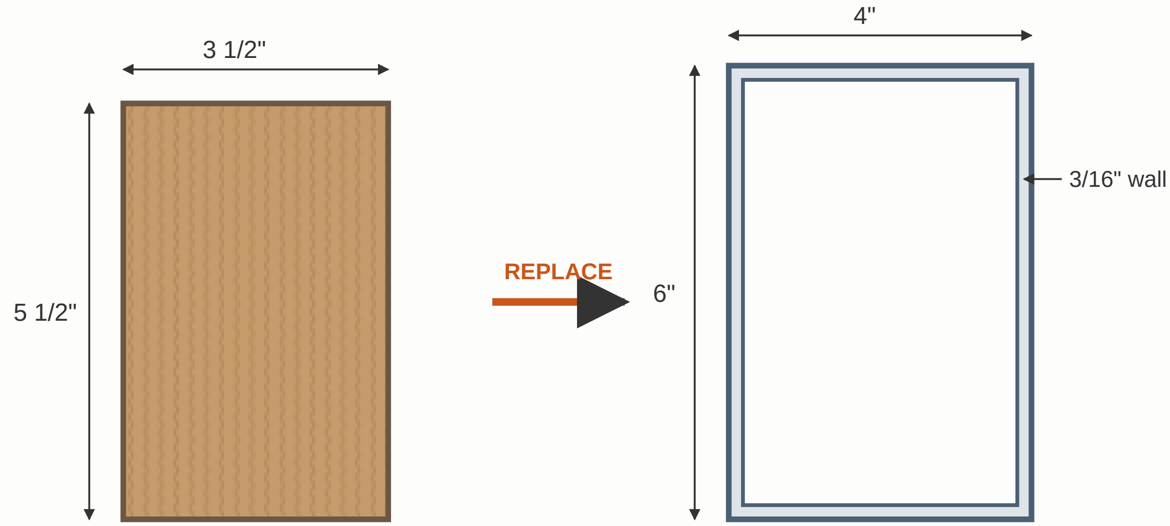
4. DRAWINGS

DWG-1 · GABLE ELEVATION — LOCATION OF V-SECTIONS TO BE REPLACED (N.T.S.)



Each "V-section" = one pair of sloped rafters meeting at the ridge. Exact members to be marked on site with owner prior to demolition.

DWG-2 · CROSS-SECTION — EXISTING WOOD vs. NEW ALUMINUM (SCALE ~1:4)



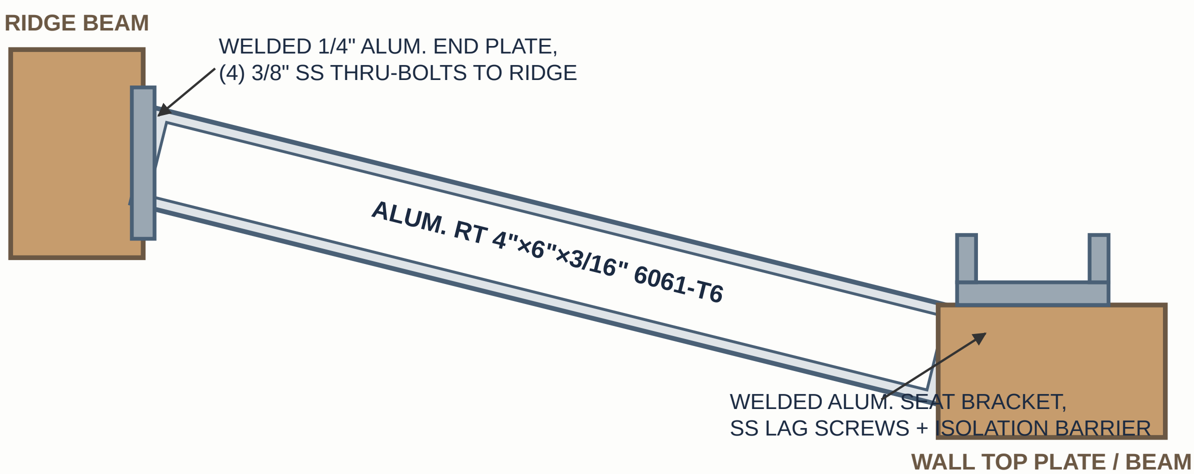
EXISTING WOOD 4×6
solid timber · 4.8 lb/ft

NEW ALUMINUM 4"×6"×3/16"
6061-T6 tube (EMS) · 4.2 lb/ft · finish to match wood

Strength: 6.7× bending capacity, 2.7× stiffness of existing timber.

Alternate size available: 3"×5"×3/16" 6061-T6 (slightly slimmer profile, still 4.2× wood bending strength, 1.4× stiffness).

DWG-3 · TYPICAL REPLACEMENT MEMBER — CONNECTION DETAIL (N.T.S.)



All welds per AWS D1.2. Stainless hardware. Aluminum isolated from treated lumber (bituminous tape / neoprene) to prevent galvanic corrosion.

Final connection design to match existing geometry; brackets shop-welded, finish-coated after fabrication.

5. PRICING

Item	Description	Amount
1. Mobilization, protection & temporary shoring	Ladders/scaffold, floor & glazing protection, shoring	$ [____]
2. Demolition & disposal	Remove (2) wood V-sections, haul & dispose	$ [____]
3. Aluminum material — Eastern Metal Supply	RT 4"×6"×3/16" 6061-T6, plates, hardware	$ [____]
4. Finish	Powder coat / finish to match existing timber color	$ [____]
5. Fabrication & welding	Cut, weld brackets & plates, drill (AWS D1.2)	$ [____]
6. Installation	Set members, SS fasteners, isolation, sealants, cleanup	$ [____]
	Subtotal	$ [____]
	Sales tax ([%])	$ [____]
	TOTAL	$ [____]

6. TERMS & CONDITIONS

- **Payment:** [50%] deposit to schedule and order material; balance due upon completion and owner walkthrough.
- **Schedule:** Approx. [X] weeks material lead time + [X] days on site, weather permitting.
- **Field verification:** All lengths, pitches and connection geometry will be field-measured before fabrication.
- **Warranty:** One (1) year on workmanship. Material per manufacturer/supplier warranty.
- **Exclusions:** Building permit & engineering letter (available as add-on if required by the county), repair of concealed rot/damage in members not listed, roofing repairs beyond the work area, painting of existing wood, electrical work.
- Any additional work will be quoted and approved in writing before proceeding (change order).

JRG Welding Corp — Authorized Signature / Date

Client Acceptance — Signature / Date